

E84 Hardware Circuit Testing

Team members and individual responsibilities:

Name:	Responsibility:
Andrew Williams	- Simulate the circuit - Build circuit on breadboard - Calculate duty cycle - Proofread write-up - Complete writeup
Wilson Zambrano	- Gather material - Simulate the circuit - Build circuit on breadboard - Run analysis - Complete writeup - Build Schematic

Hardware:

Describe what you intend the circuit to do, draw the circuit diagram, and why you selected what values for the components you chose. Please include a diagram of your circuit, an image of the circuit, as well as an image of the oscilloscope with the input and output voltages. Explain your initial results. If your circuit doesn't output what you expected from your analysis and simulation, think of some reasons why it didn't work. For your final report next week, try to redesign the circuit so it works.

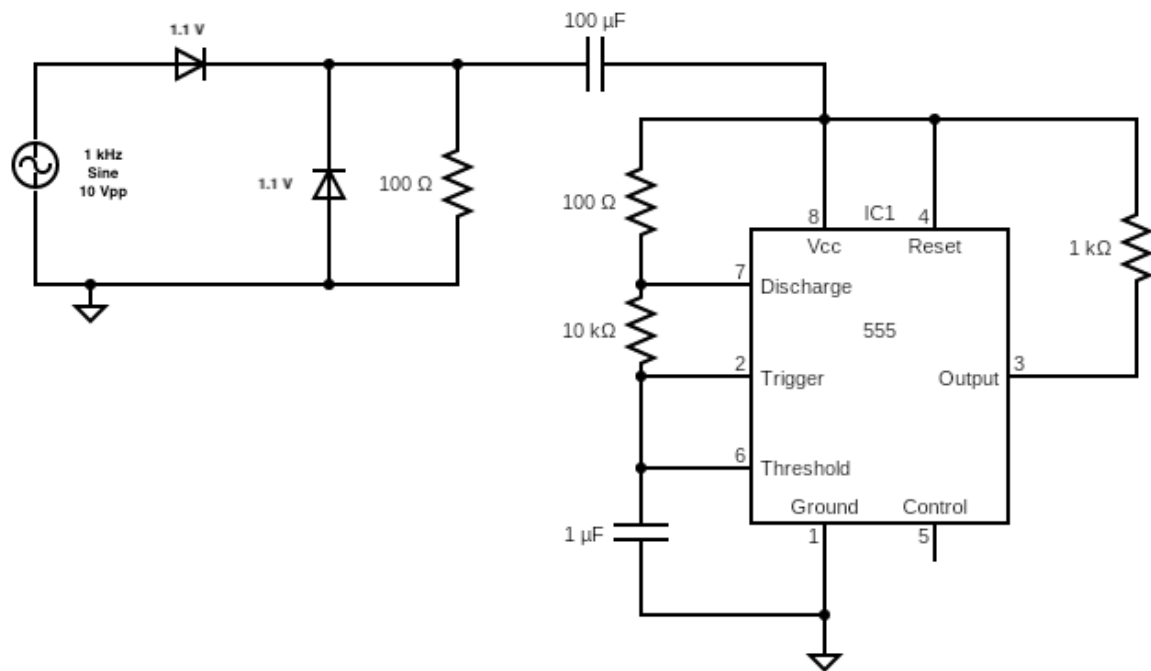


Fig. 1: Circuit Schematic

The circuit shown in Fig. 1 is designed to convert AC power into DC power through a series of stages. The first stage is a low pass filter which attenuates the high frequencies and allows the low frequencies to pass through. In practice, an AC signal with a very small frequency and amplitude approximates to be DC power.

In the selection of component values for the low pass filter 3 factors were considered: capacitance, resistance, and diode. A high enough capacitance was preferred to ensure that the capacitor has a sufficient discharge period when the output voltage drops below the AC voltage. The time constant ($R \times C$) guided the selection of resistors and capacitors. A high enough resistance value was preferred to ensure that the discharge occurs very slowly, prolonging the discharging period. A diode was

oriented in the forward direction, such that the circuit is only able to emit the positive forward voltage, effectively eliminating any negative voltage.

The second stage incorporates a 555 timer which generates a series of pulses. These pulses are intended to control the pulse width modulations of the waves.

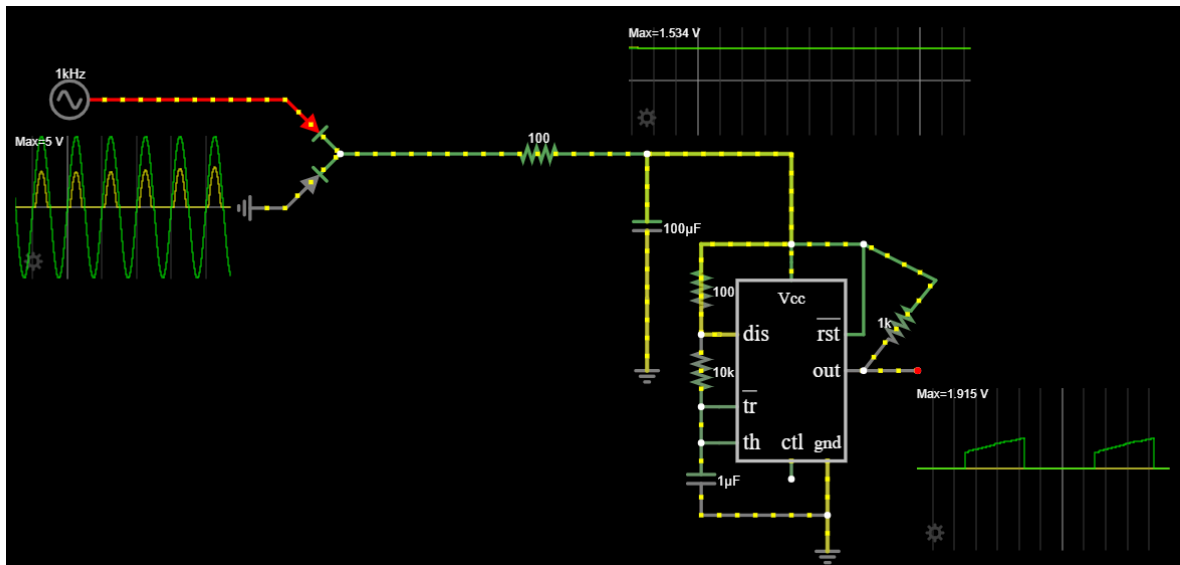


Fig 2. Falstad Simulation of the Circuit

As depicted in Fig. 2 shows a waveform generator inputs a 5 peak to peak voltage (V_{pp}) sine wave with a frequency of 1kHz. This input is then fed into a diode with a forward voltage of 1.1V. The conversion from alternating current to direct current is achieved using a capacitor of sufficient size, which charges and discharges at a slow rate when paired with a large resistor.

Given that this circuit is a half-wave rectifier, only half of the 10V_{pp} input, or 5 V_{pp}, is allowed to pass through and form the output. This is inherent to the process of half-wave rectification, which permits only one half (the positive or negative half depending on the orientation of the diode) of the input wave to pass through. Consequently, the output is a series of positive (or negative) half-cycles of the AC wave, effectively converting the AC input into a pulsating DC output.

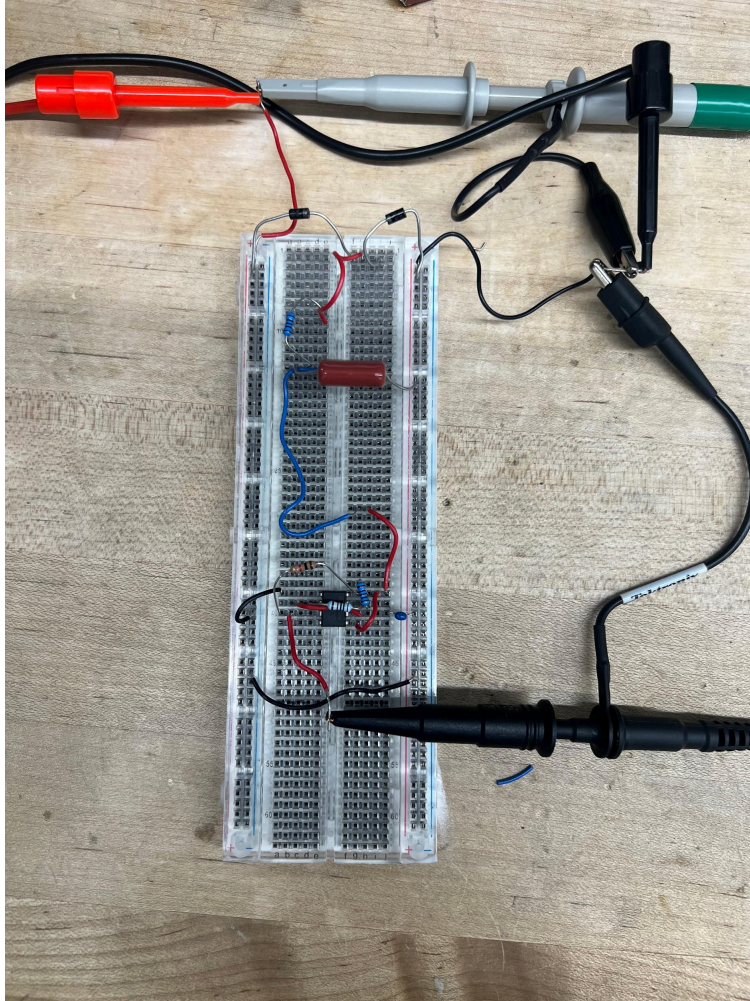
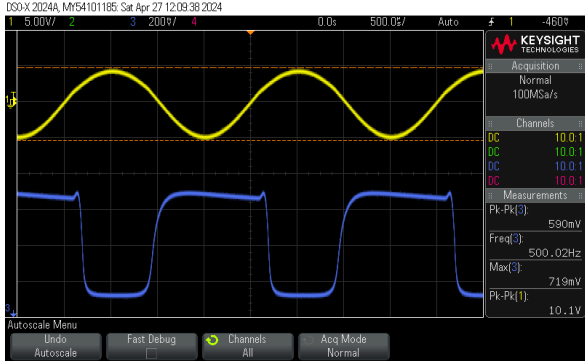
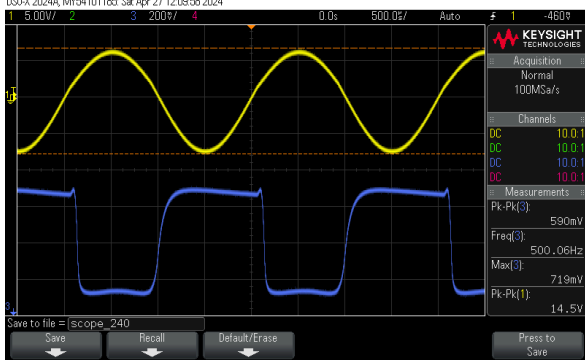
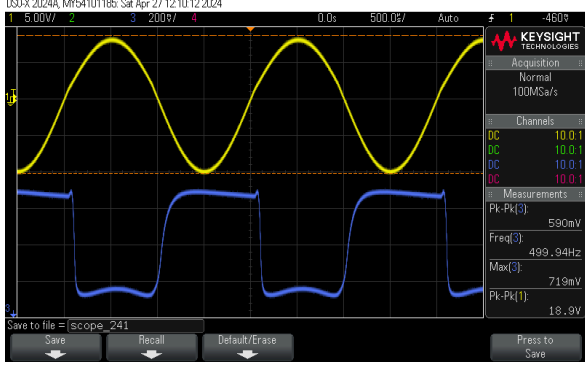
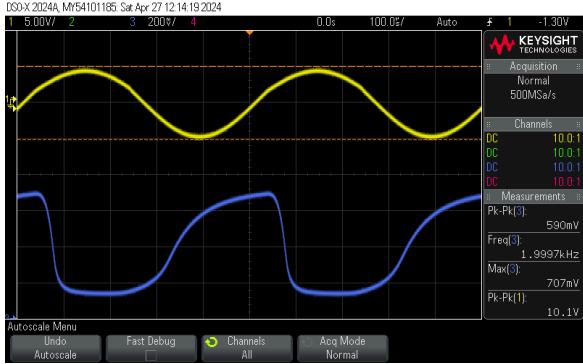
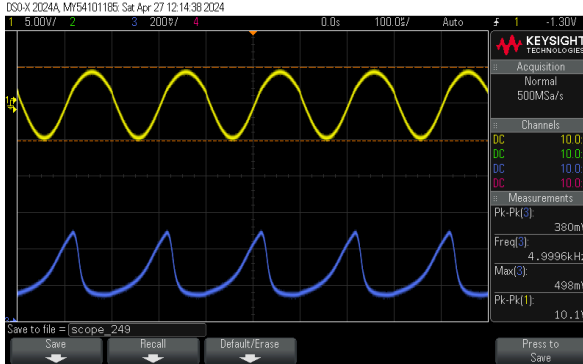


Fig. 3: Hardware Circuit Picture

Fig. 3 represents the circuit built on the basis from the schematic shown in Fig. 1, that is used to determine the duty cycles by altering the V_{pp} and the frequency of the alternating current source.

Peak-Peak VAC (V_{pp})	Frequency (Hz)	Duty Cycle (%)	Oscilloscopes (Yellow = AC, Blue = PWM)
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10	500	60.78431373	 Fig. 7
15	500	56.07843137	 Fig. 8
20	500	51.76470588	 Fig. 9

10	2k		 Fig. 16
10	5k		 Fig. 17

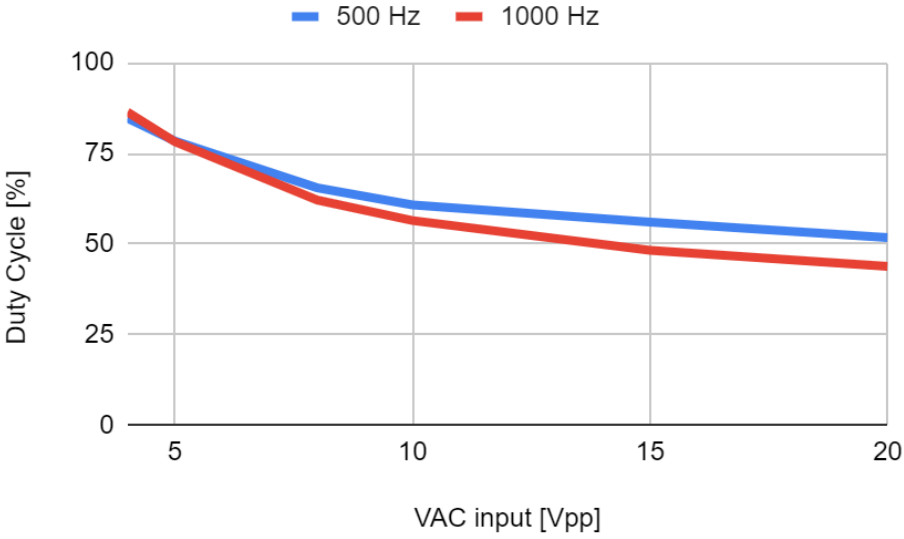


Fig. 18: Plot of the output Duty Cycle vs. AC signal voltage.

Based on the results, the difference of 500 Hz in frequency has a low effect on duty cycle. Changing AC voltage amplitude from a range of 2 to 10 V has a much greater effect on the duty cycle. Above

about 8 V_{pp}, the max output voltage plateaus at about 720 mV. Thus, we have succeeded in making a circuit that makes duty cycle proportional to VAC amplitude, but does not affect the output amplitude.

Since the low pass filter is integrated with a half wave rectifier, it can be witnessed in some figures (namely, Fig. 4, Fig. 11, Fig. 16 and Fig. 17) how the diode is discharging and charging up to maintain positive voltage. Taking a look at low frequencies, as shown in Fig. 4, it behaves like a sine wave. At high frequencies just like in Fig. 16 and Fig. 17, the circuit reverts and behaves like a sine wave. However, taking a look at Fig. 11, some intermediate frequency, it is clear that there exists some sweet spot at which it behaves a lot like a step function. Thus, future experiments should consider at which frequencies produce the flattest peak. This could have a possible application for charging appliances with limited power. If there are situations where it is better to limit incoming voltage as a duty cycle with higher amplitude rather than a lower DC voltage, this circuit can be used. However, in applications where higher voltages are preferred, it would be logical to include transformers into the circuit.



Fig. 19: Process of Finding the Duty Cycle

Fig. 19 shows the process of determining the duty cycle which includes uploading the screenshot from the oscilloscope and dividing the number of pixels that the circuit produced a positive signal over the total number of pixels in the period.

[Discussion of error]

- There may be impedances from the wires
- The process of finding the duty cycle is vulnerable to systematic error in calculating the number of pixels that has a peak over the total number of pixels in that period.
- It was difficult to measure the duty cycle for some of the figures. The step changes were sometimes too smooth to concretely distinguish where the step begins and ends. We measured around the vertical center of the plots. If this project were to continue, we would experimentally and computationally (through circuit analysis) determine what frequency creates the “sharpest” edges.
- Tolerances on the resistors range from 0-5%.